

Economic Complexity Research Proposal: Large Jumps in the Product Space

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1. Research problem

Understanding how and why countries diversify into new products is crucial for designing effective industrial and innovation policies. Studying the dynamics of these moves can reveal the mechanisms that enable some nations to successfully upgrade their economies while others struggle. Countries are more likely to move towards related product categories in the product space, because of the shared capabilities and knowledge, that makes it easier to succeed compared to greatly differing product categories. (Hidalgo, Klinger, Barabási, & Hausmann, 2007). This tendency reflects the notion of path dependency in economic development, where existing capabilities shape and constrain future diversification, making related products easier to adopt and reducing the risk of failure compared to distant jumps in the product space.

However, some countries attempted to enter high complexity products far from their capability set, with mixed success. As an example, South Korea, despite a relatively low-complexity industrial base in the 1970s–1980s, made a long-distance move into semiconductors, ultimately developing a globally competitive memory-chip industry. Bangladesh, on the other hand, has attempted to move beyond low-complexity garments into more complex products such as pharmaceuticals and electronics/electrical machinery, but these entries have generally faced capability and scale constraints.

The aim of this research would be to answer the following questions:

1. Under what conditions do countries attempt to enter far-distance product categories, moving beyond their current capabilities?

2. How likely are these entries to survive, compared to entries related to existing capabilities? (We know beforehand that it is lower, but how much?)
3. What are the indicators that predict survival, meaning a successful long-term entry to a product category?

2. Data

The backbone of the data would be the economic complexity index (ECI) and the product complexity index (PCI) and its relatedness network from the OEC website.

The other independent variables are various financial, economic and socio economic data sources, which have the potential to be significantly influential. The following list contains the possible variables, some with previous scientific foundations, others by intuition and that they have a reason to be potential candidates. These variables may be subject to change.

Candidate variables:

- Institutional quality

Strong institutions provide long term stability for investment, transparency and contract enforcement. Weak institutions cause delays, corruption, supply chain disruptions, and uncertainty. Complex products expectedly require more of a stable environment. One study “concludes that institutional quality enables economic complexity to spur less reliance on natural resources and pave the way to evade resource curses in resource-rich African countries.” (Olaniyi & Odhiambo, 2025)

For this variable the Worldwide Governance Indicators (WGI) seem a right choice.

- Human capital

High human capital seems unavoidable to introduce new products, that have a requirement for various capability sets, that are currently not present. High human capital also allows

flexibility and easier retraining into the required sector. This fundamental idea is also supported by previous scientific research. (Leite & Cardoso, 2022)

To capture this, a potential dataset is the Barro-Lee Educational Attainment Dataset, containing worldwide education attainment data.

- FDI

Foreign direct investment can be the right tool that bridges the gap between the current and the needed capabilities, even though certain capabilities and knowledge are tacit and not easily bought. Research shows that inward FDI per capita Granger-causes economic complexity, although with some caveats. (Antonietti & Franco, 2021)

The UN Trade & Development website, has a reliable dataset for foreign direct investment on inward and outward flows.

- Participation in trade agreements

Trade agreements can facilitate market access, which naturally allows countries to export more of a certain product increasing its RCA. They can have other less direct effects on a country, including easier technological transfer due to integrating more into the global supply chain and compliance with international standards.

This could be measured as dummy variables, whether a country is or is not a part of a certain major trade agreement, like EU, ASEAN or USMCA.

- International migration

Strong migration and diaspora networks can support economic complexity because migrants bring tacit knowledge, business connections, and cultural-institutional capital that help diversify a country's productive capabilities. As Bahar, Rapoport, and Turati (2022) find, countries with higher *birthplace diversity* among immigrants are significantly more economically complex.

The UN DESA International Migrant Stock dataset can be used, which provides estimates of migrant stock by origin and destination.

It would also be beneficial to examine how the exact product categories affect the success rate. However this may not fit in the scope of the analysis, but this can be decided during the implementation of the research.

3. Methodology

Step 1: Identifying product entries

For each country, detect product appearances, by finding products with $RCA \geq 1$, where this condition doesn't hold for $t-1$.

Extra consideration needs to be taken for filtering out instances, where an RCA value hovers around 1.

Step 2: Measuring entry distance

Measuring the minimum distance of the new entries to their closest already established product categories (with $RCA \geq 1$). From this, we can identify whether the new entrance is a large jump or a closely related product category. This can either be a binary value with a clear cut-off, or we can treat it as an integer variable.

Step 3: Checking entry survival

Measuring whether $RCA \geq 1$ for the following years. It is advised to check for multiple years, to avoid fluctuation around $RCA = 1$ here also. With this we can identify whether a product category entry survived or failed to establish itself long term.

Step 4: Cleaning and integrating variables into a common data frame

Due to collecting data from different sources, careful consideration must be taken, that the data is useful and reliable for all categories, without a significant number of NA values. This will probably limit the countries and the timeframe where a model can be applied.

Step 5: Applying binary regression model

To determine, what kind of conditions affect the survivability of an entry into the product category we can apply a binary model, like logistic regression, where we include the potentially influential variables, and control for the calculated distance.

Using backward elimination we can create a model containing only the significant variables, then including interaction terms with distance to identify how the effect of the other variables change with distance.

This complicated model, however, would benefit from strong additional measures to ensure robustness.

4. Expected outcomes

The expectation of the outcome is first to quantify how much distance, for the closest node of the product category that a country is already specialized in, affects successful entries. This is expected to match previous literature, that higher distance should greatly decrease successful entries.

A less trivial expectation is that we can successfully identify several conditions that make entries into product categories long term successful. Pinpoint these areas will help us understand what social, economic and political conditions help or inhibit the successful survival of new entries.

Furthermore, we expect that distance in the product space significantly impacts most of the conditions for successful entry. If the conditions needed wouldn't change with distance, it would mean that it is simply a smaller chance, if distance increases. However, we await that there are certain conditions for large jumps in the product space that are different from entering into a neighbouring product category.

5. Impact

This research can reveal pathways for low-complexity countries to overcome path dependence and escape reliance on commodity exports. By identifying possible large jumps, it can support economic diversification, increasing resilience and long-term growth potential, which directly benefits citizens through higher social welfare and employment opportunities.

For investors and firms, the research provides guidance on which markets are most likely to yield successful ventures and which carry higher risk, improving resource allocation and reducing the likelihood of loss.

Overall, the study offers strategic insights for both public and private actors, aligning economic complexity theory with practical decisions that can enhance growth, resilience, and inclusive development.

6. Use of AI

In this research proposal AI was used to help brainstorm the topic itself. Furthermore it was used to search for related scientific literature and suggest possible variables for the statistical model. AI wasn't the only approach applied for these tasks. The tool was also used for rephrasing and restructuring to achieve better readability.

7. References

Antonietti, R., & Franco, C. (2021). *From FDI to economic complexity: A panel Granger causality analysis*. *Structural Change and Economic Dynamics*, 56, 225–239. <https://doi.org/10.1016/j.strueco.2020.11.001>

Bahar, D., Rapoport, H., & Turati, R. (2022). *Birthplace diversity and economic complexity: Cross-country evidence*. *Research Policy*, 51(8), 103991. <https://doi.org/10.1016/j.respol.2020.103991>

Hidalgo, C. A., Klinger, B., Barabási, A.-L., & Hausmann, R. (2007). *The product space conditions the development of nations*. *Science*, 317(5837), 482–487. <https://doi.org/10.1126/science.1144581>

Leite, D. W., & Cardoso, L. C. B. (2022). *Human capital and technology in the growth of economic structure*. *Investigación Económica*, 82(323), 27–52. <https://doi.org/10.22201/fe.01851667p.2023.323.82953>

Olaniyi, C. O., & Odhiambo, N. M. (2025). *Does economic complexity provide antidotal pathways to evade the resource curse syndrome? A novel role for institutions*. *Sustainable Development*, 33(2), 3118–3142. <https://doi.org/10.1002/sd.3266>